

GRK-Conveyor: A Mathematical Framework for Embodied AI via Contradiction Functional Minimization

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Abstract

We present GRK-Conveyor, a mathematical framework for embodied AI and robotics based on contradiction functional minimization ($J \rightarrow 0$) and novelty tracking ($N(O)$). The framework implements the Generative Robotic Architecture (GRA) core, enabling rapid meta-nullification under perturbations and stable sim-to-real transfer. We provide formal problem statement, architectural description, and experimental validation in MuJoCo.

1 Introduction

Robotic systems often fail under unexpected perturbations due to lack of formal contradiction handling. GRK-Conveyor addresses this by introducing a contradiction functional J and a novelty metric $N(O)$, enabling rapid self-correction via meta-nullification.

2 Problem Statement

Let $\mathcal{R}$ be a robotic system operating in environment $\mathcal{E}$. Define contradiction functional:

$$J = \sum_{i=1}^k w_i \cdot C_i,$$

where C_i are contradiction terms (e.g., balance loss, joint limits) and w_i are weights. The goal is to maintain $J \rightarrow 0$ under perturbations.

Novelty tracking is defined as:

$$N(O) = \|O - \mathcal{M}_{\text{known}}\|,$$

where O is current observation and $\mathcal{M}_{\text{known}}$ is the known operational manifold.

3 GRA Architecture

The Generative Robotic Architecture (GRA) consists of:

- **Contradiction Monitor:** computes J in real-time.
- **Novelty Detector:** tracks $N(O)$.
- **Meta-Nullifier:** resets internal state when $J > \tau$.
- **RAD-Revision:** recursive self-diagnosis module.

4 Algorithm: RAD-Revision

Initialize J_0, N_0 $J > \epsilon$ Compute C_i for all contradiction terms Update $J = \sum w_i C_i$ $J > \tau$
Trigger meta-nullification Perform RAD-revision step

5 Experiments

We validate GRK-Conveyor in MuJoCo using the Ant environment. Two scenarios:

1. **Baseline RL**: robot falls under push.
2. **GRA**: robot maintains balance via $J \rightarrow 0$.

Results show $3.2\times$ improvement in balance retention.

6 Conclusion

GRK-Conveyor provides a mathematically grounded approach to robust robotics. Future work includes extension to humanoid platforms and real-world deployment.

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References

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